LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to other reference systems viz.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to other reference systems viz. The national grid system, the universal transverse Mercator grid and the geographical reference system.
5. Specific Objectives	14.1.1.1 Trainee will be able to explain and define a position on the earth.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of map projections and coordinate definitions; Safety: Handle maps carefully; Supervision: Instructor monitors understanding of grid zones.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of a reference system?
(b) How is a position defined on a standard map?
(c) Why are different grid systems required for navigation?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Geographical Reference System definition • Latitude and Longitude coordinates • Prime Meridian and Equator roles • Global positioning standards	L & Ds	PPT, IB	15 min
(b)	• National Grid System overview • Universal Transverse Mercator grid • Zone division and coordinates • Grid convergence explanation	L & Ds	PPT, PDF	20 min
(c)	• Comparing grid systems • Application in aviation contexts • Accuracy and precision factors • Selection criteria for systems	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between geographical and national grid systems.
(b) How does the Universal Transverse Mercator grid improve navigation accuracy?
(c) What factors influence the choice of a reference system for a specific mission?

12. Home Assignment and Follow Up	Prepare a brief note comparing the National Grid System and the Universal Transverse Mercator grid, highlighting their specific applications in aviation.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to other method of expressing positions i.e. pin points, range and bearing.	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min